TrES-1b

R-0220

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_250614` · created 2026-07-17T01:26:51+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460840.7688 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1619 +/- 0.0093
Transit depth (Rp/Rs)^2	2.62 +/- 0.3 [%]
Orbital Inclination (inc)	87.07 +/- 0.73
Airmass coefficient 1 (a1)	1.015 +/- 0.011
Airmass coefficient 2 (a2)	-0.0118 +/- 0.0092
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#1 - [515, 397]
Optimal Method	PSF photometry
Transit Duration (day)	0.093 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1619 ± 0.0093 vs 0.1358 (deviation 2.81σ)
Duration fit vs published	0.093 d vs 0.1042 d (deviation 1.67σ)
Mid-time O–C	7.7 min
Depth SNR	17.4
Residual scatter	1.08 %

Light curve

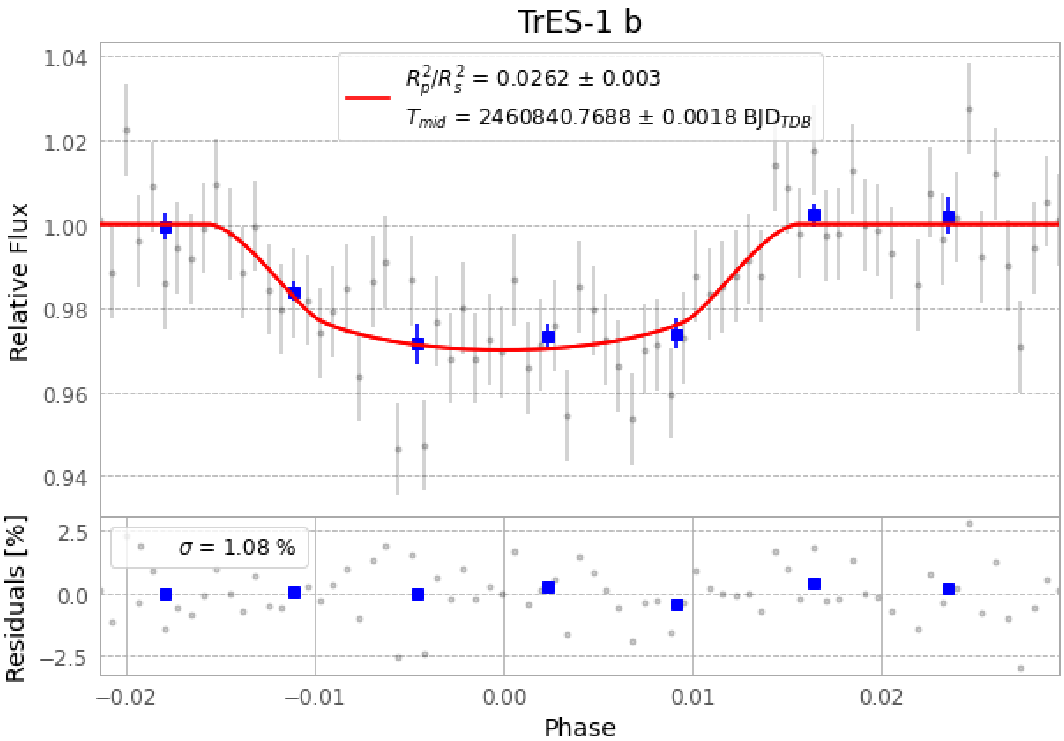

Figure 1 — FinalLightCurve_TrES-1 b_2025-06-13.png (SHA-256 f28ff2b0403bef60…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 215], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b311f8377864475f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

Data provenance

75 FITS frames (49 MB), `TRES-1250614044815.FITS` → `TRES-1250614083015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250614.log (a314e5a8818c…), FinalLightCurve_TrES-1 b_2025-06-13.png (f28ff2b0403b…), FinalLightCurve_TrES-1 b_2025-06-13.csv (64b76c00ba7e…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)